

Retail Customer Behavior & Sales Analytics: Business Problem Statement

Overview Retail businesses face a constant balancing act between inventory costs and sales margins. The core challenge for this retail operation is determining how to optimize vendor relationships, price products effectively, and reduce the financial burden of unsold, slow-moving inventory.

To solve this, the analysis is driven by 7 core business questions designed to uncover insights across procurement, capital efficiency, and pricing strategy.

Core Business Questions:

- 1. Vendor Profitability** Which vendors are the most profitable based on gross margin and total sales? *Goal: Identify the highest-performing suppliers to prioritize relationship management and negotiate better terms.*
- 2. Pricing Relationship** Is there a relationship between the purchase price and the sales price of items? *Goal: Understand standard markup practices and identify if certain pricing tiers yield better margins.*
- 3. Sales Velocity** Do higher-priced items sell faster or slower than lower-priced items? *Goal: Determine product price elasticity and adjust inventory orders based on how fast different price brackets move off the shelves.*
- 4. Vendor Dependency** Which vendors have the highest and lowest purchase contribution? *Goal: Analyze where the business is spending its money to identify over-reliance on single vendors or opportunities to consolidate buying power.*
- 5. Capital Efficiency** Which vendors have low inventory turnover, indicating excess stock and slow-moving products? *Goal: Pinpoint operational inefficiencies where warehouse space and capital are being wasted on products that do not sell quickly.*
- 6. Trapped Capital** How much capital is tied up in unsold inventory, and how does this affect overall profitability? *Goal: Quantify the financial impact of over-ordering and identify clear targets for inventory reduction and discounting strategies.*
- 7. Brand Positioning** Is there a difference in sales performance between premium brands and budget brands? *Goal: Evaluate whether the store's customer base is driven more by volume (budget) or margin (premium) to guide future marketing and purchasing decisions.*